5. Summary

I began by loading the dataset and examining it with head(), shape, info(), and describe(), which revealed that numeric features like Sales and Profit exhibited right-skewed distributions—with means significantly higher than medians—indicating the presence of high-value outliers. Through histograms and boxplots, I confirmed this skew and identified extreme values. Weekly time-series plots uncovered seasonal patterns and periodic dips. Bivariate plots (box, violin, point, and count plots) revealed regional disparities in profit and highlighted class imbalances—particularly the dominance of ‘Furniture’. During preprocessing, I removed duplicates, handled missing values through row deletion or median imputation, and eliminated outliers using the IQR method, which reduced skewness while preserving the majority of data. I converted date fields to datetime, categorical text fields to category, and applied one-hot encoding to nominal variables, preserving numeric consistency and avoiding ordinal misinterpretation. Temporal features like week, month, and year were extracted, and numeric columns were standardized. I then investigated feature redundancy via a correlation matrix and heatmap, which revealed a high correlation ($r \approx 0.9$) between Sales and Profit; to reduce multicollinearity, I dropped one of these. The result is a clean, lean dataset with encoded categorical features, scaled numeric variables, no missing values or duplicates, and minimal multicollinearity—well-prepared for robust model training and evaluation.